Teaching Statement

After teaching Calculus I lab sections from August 2020 to May 2022, I reached a local maximum in my instructional journey when I became Instructor of Record in Fall 2022. After a very successful semester of teaching College Algebra, the department saw my high student engagement and positive reviews and assigned me the business applied algebra course "Finite Math." This was surprising to me, as graduate students are not usually assigned to teach this course. Finite is known as the "weeder" course for business majors, and thus the general apathy toward math compounds into a challenging classroom setting. To address this early, on the first day I instructed my students that they must take their education into their own hands. I frankly told them I was a graduate student, and whether they passed or failed the course was entirely up to them. This blunt message was obviously met with lower student engagement with me as an instructor, as well as negative course evaluations. I have since learned to embrace a more cooperative educational approach, rather than an authoritarian stance. Since that day, this more caring approach has led to continual improvement in my teaching style, as well as in student performance. Whether it's scheduling meetings with struggling students, or emailing kudos for a job well done on an exam, a student-forward approach has become key to my teaching philosophy. Nowadays, my Finite math students say that my teaching style "made a subject that I am not particularly great at one of my favorite classes to attend. [Mr. Tuttle] teaches in a way that connects with his students and makes it clear that he truly cares about educating every single individual in his class."

However, simply caring for students does not automatically make a great educator. Often as teachers, we are so preoccupied with seeking to instruct students that we forget that we too must be continually learning. For example, the first step I take in learning about my students is to provide them with a survey at the start of the semester. I ask questions such as what learning practices have gone well in the past for the student, as well as what tactics teachers have tried that they like or dislike. Using this information, I design lesson plans that work with and for my students, rather than independent of my students. Later, I follow up with a mid-semester survey to gather further feedback and then implement any changes or extra support for the students well before the course comes to a close.

One semester I had a student express to me that they historically were a little slower than their peers at grasping new concepts. After many discussions and office hours assisting and listening to the student, I discovered that taking the time to write explanations in lecture, rather than just speaking them aloud, helped this student succeed. Accommodating this student's difficulty processing verbal explanations was well received by the entire class. Ever since then I have written out more of the class polling statements and used a virtual whiteboard during class. This strategy provides students who have trouble processing and note-taking another resource to look back on later to complete their notes and examples. Coming to this realization required effort and individual communication outside the classroom. Indeed, the sections I teach are larger, usually 60 to 90 students. Despite this, we as instructors have a duty to not simply lecture

and leave, but to connect one-on-one with our students. From this effort, I've found many of my non-STEM majors' opinions regarding math have changed from scared, or hateful, to something more neutral: "As for people that do not enjoy math, I had multiple classmates that struggled heavily to succeed and they said because of [Mr. Tuttle] their outlook on math changed entirely." This shift in perspective is my daily goal.

Students succeed in an environment that incorporates various learning approaches. For example, visual tools were central to my Calculus II course last summer. During the solids of revolution unit, I coded a Wolfram notebook of 3D diagrams of the solids we were building, incorporating visible slices of the shape over which we integrated. Approaches such as these led a student who had previously failed the class to write in their evaluation: "I actually look forward to listening to lecture. I never dread going to class even as a commuter during the summer." I also incorporate active learning into my courses. These include "think pair share" activities on possible outcomes of word problems or polling-style review sessions that engage participants.

In a subject as theoretical as mathematics, tactile learning has a persistent place. I had a precalculus teacher expose students to radians via a hands-on activity using yarn to the length of various radii of circles. I hope to use a similar activity when I teach precalculus in the future, and took inspiration from it to design an activity for my current Finite Math course. After watching a pre-class introduction video, students worked in groups on a task involving ranking Skittles flavors and relating the concept to factorials and permutations. The activity turned one of the most intimidating lessons of the year into a memorable, joyful activity.

In a service-based capacity, I volunteer at the University of Arkansas Math Discovery Day outreach event each year. Geared toward youth ages 6 to 16, various booths and math games are set up in the local library, exposing participants to a variety of topics in a hands-on manner. I lead the Platonic Solids and Gaussian Curvature booth. Utilizing large models of Platonic Solids, we expose students to Euler characteristic, as well as polygon angles in non-Euclidean geometry. Participants also draw on a map the path they think will be quickest between two distant cities. We then copy the path onto a globe and use a string with magnets to show the correct shortest path, which looks curved on a flat map. This combination of visualization and tactile learning has been extremely successful, particularly with budding young mathematicians.

Mentorship is an important part of being a professor. We are here not only to teach math, but also to help students grow as humans. Beyond the classroom setting, I hope to mentor new mathematicians through undergraduate research projects. As part of my dissertation, I developed Python code to calculate Legendrian Contact Homology of knots in special manifolds. In a less hyper-specific setting, knot theory is very accessible to undergraduates. Thus I envision research projects calculating knot invariants as a central part of my continued research, as they expose students to both topological and algebraic concepts with an algorithmic flavor.

Whether it is in a research setting, a lecture hall, or a library full of curious minds, my goal is the same: to challenge students critically all while making sure they feel valued along the way. On my first day of Finite, I told students their success was up to them. Now, together we cultivate the seeds of success, and watch the bounty grow.